SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1701**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.*(BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 3 **Duration : 1 Hour**
Total Marks:30

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

*I **GIFTSON KEVIN MATTHEW.Y (192511167)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: GIFTSON KEVIN MATTHEW.Y

Annexure A

Constraint-Based Problem Solving

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that:

   Constraints:

   i. D1 cannot work Night shift
   ii. D2 must work before D3 (Morning < Afternoon < Night)
   iii. Only one doctor per shift
   iv. D3 cannot work Morning
   v. All doctors must be assigned exactly one shift

   Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

   Constraints:

   i. Movement allowed: Up, Down, Left, Right
   ii. Cost of each move = 1
   iii. Cannot pass through obstacles
   iv. Use Manhattan Distance heuristic

   Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost.

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information.

The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

Constraints:

- The robot can move up, down, left, right
- Some cells are blocked (obstacles) and cannot be traversed
- The robot has limited sensing capability (can only observe adjacent cells)
- Each movement has a cost of 1, but entering risky zones has an additional cost of 2
- The robot must minimize total cost while ensuring safe navigation
- The robot must update its knowledge dynamically (online search scenario)

Tasks:

a) Analyze all the given constraints and explain how they affect the robot's decision-making process.

b) Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved.
c) Demonstrate how the robot applies AI concepts such as:

- Intelligent agents
- Rationality
- Environment type

d) Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

1. Hospital Doctor-Shift Assignment Using Backtracking Search

Given:

There are three doctors:

- D1
- D2
- D3

There are three shifts:

- Morning
- Afternoon
- Night

Constraints:

1. D1 cannot work in the Night shift.
2. D2 must work before D3.
3. Only one doctor can be assigned to each shift.
4. D3 cannot work in the Morning shift.
5. Every doctor must be assigned exactly one shift.

The problem can be solved using Backtracking Search. In this method, we assign a shift to a doctor and check the constraints. If any constraint is violated, we go back and try another possibility.

Step 1: Assign D1

Possible shifts for D1 are:

- Morning (OK)
- Afternoon (OK)
- Night (ON)

D1 cannot work at Night.

So first, take:

D1 → Morning

Now the remaining shifts are Afternoon and Night.

Step 2: Assign D2

D2 has to work before D3.

Since D1 is already in Morning, D2 cannot be in Morning.

Try:

D2 → Afternoon

Now D3 must work after D2.

The only remaining shift is Night.

Step 3: Assign D3

Try:

D3 → Night

Check the constraints:

- D1 is not in Night (OK)
- D2 is before D3: Afternoon < Night (OK)
- Only one doctor in each shift (OK)
- D3 is not in Morning (OK)
- All doctors have exactly one shift (OK)

Therefore, this assignment is valid.

Backtracking Check

We can also see the invalid possibilities:

Assignment	Result
D1 → Night	Invalid, because D1 cannot work Night
D3 → Morning	Invalid, because D3 cannot work Morning
D2 → Night, D3 → Afternoon	Invalid, because D2 must work before D3
D1 → Morning, D2 → Afternoon, D3 → Night	Valid

Final Valid Schedule

Shift	Doctor
Morning	D1
Afternoon	D2
Night	D3

Therefore, the final valid assignment is:

D1 → Morning, D2 → Afternoon, D3 → Night.

Thus, Backtracking Search successfully finds a solution by checking the constraints at every step and rejecting assignments that violate the conditions.

2. Robot Grid Problem Using BFS and Manhattan Distance

The robot is working in a 5 × 5 grid. It has to move from Start (S) to Goal (G).

The given constraints are:

- Movement is allowed only Up, Down, Left and Right.
- Cost of each movement is 1.
- Obstacles cannot be crossed.
- Manhattan Distance is used as the heuristic.

The Manhattan distance is calculated as:

$h(n) = |x - xg| + |y - yg|$

where (x,y) is the current position and (xg,yg) is the goal position.

Search Approach

For each node, the robot checks its possible neighbouring cells.

The possible movements are:

1. Up
2. Down
3. Left
4. Right

If a cell is an obstacle, it is not added to the search.

The cost of reaching a node is:

$g(n)$ = number of moves made

The Manhattan distance gives the estimated distance to the goal:

$h(n) = |x-xg| + |y-yg|$

Therefore, when heuristic information is considered, the evaluation value is:

$f(n) = g(n) + h(n)$

Step-by-Step Procedure

Step 1: Put the starting position S into the search structure.

Step 2: Expand S and find all valid neighbouring cells.

Step 3: Remove cells that are outside the grid or are obstacles.

Step 4: Calculate the cost and Manhattan distance for every valid cell.

Step 5: Add the valid cells to the priority queue according to their evaluation value.

Step 6: Expand the most suitable node and repeat the same process.

Step 7: Continue until the Goal G is reached.

Step 8: Trace the parent nodes from G back to S to obtain the final path.

Example of Node Evaluation

For a node (x,y):

g(n) = cost from S to current node

h(n) = |x-xg| + |y-yg|

f(n) = g(n) + h(n)

For example, if the current node is (2,3) and the goal is (4,4):

h(n) = |2-4| + |3-4|

h(n) = 2 + 1

h(n) = 3

If the robot has already made 4 moves:

g(n) = 4

Therefore:

f(n) = 4 + 3 = 7

The node is then considered based on its priority.

Priority Queue Updates

The priority queue contains the valid nodes that have not yet been expanded.

At every step:

Expand node → Check neighbours → Remove obstacles → Calculate cost → Insert valid nodes → Select next node

The process continues until G is selected.

Optimality

Since every movement has a cost of 1, the shortest path is the path having the minimum number of moves.

The exact sequence of coordinates and final numerical path cost cannot be determined from the uploaded question paper alone because the 5×5 grid showing the positions of S, G and the obstacles is not included in the document.

Once the actual grid is provided, the final answer can be written in this form:

S → [cell] → [cell] → [cell] → ... → G

Optimal Path Cost = Number of movements

Thus, the robot uses the search process to avoid obstacles and find the shortest feasible path from S to G.

3. Autonomous Rescue Robot – Constraint Analysis and Search Strategy

An autonomous rescue robot is deployed inside a disaster-affected building. Its main objective is to reach a survivor at position G starting from S.

The environment is partially observable, meaning that the robot does not have complete information about the entire building. It can only observe the cells around its current position.

a) Analysis of the Constraints

The important constraints are as follows.

1. Movement Constraint

The robot can move only:

- Up
- Down
- Left
- Right

Therefore, diagonal movement is not allowed.

2. Obstacle Constraint

Some cells are blocked. The robot cannot enter these cells.

So, whenever the robot observes an obstacle, it must avoid that path.

3. Limited Sensing

The robot can observe only adjacent cells.

This means it does not know the complete environment from the beginning. It has to discover the environment while moving.

4. Movement Cost

Every normal movement has a cost of 1.

Therefore:

Normal movement cost = 1

5. Risky Zone Cost

Entering a risky zone has an additional cost of 2.

Therefore:

Risky zone movement cost = 1 + 2 = 3

The robot should preferably avoid risky zones if there is another safe path with a lower total cost.

6. Dynamic Knowledge

The robot has to update its knowledge as it moves.

For example, initially it may not know that a particular path is blocked. After moving near that location, its sensors can detect the obstacle and update its internal knowledge.

Therefore, the robot must make decisions continuously instead of planning everything only once.

b) Suitable Search Strategy

For this problem, Uniform Cost Search (UCS) combined with an Online Search Agent is suitable.

Uniform Cost Search is useful because the movement costs are not always the same.

A normal cell costs:

1

A risky cell costs:

3

Therefore, simply finding the path with the fewest steps may not give the minimum-cost path.

Steps of the Approach

Step 1: Start at position S.

Step 2: Observe the neighbouring cells using the robot's sensors.

Step 3: Identify whether each neighbouring cell is free, blocked or risky.

Step 4: Calculate the cost of moving to each available cell.

Step 5: Add the possible paths to the priority queue.

Step 6: Select the path with the lowest accumulated cost.

Step 7: Move to the selected cell.

Step 8: Observe the new surroundings.

Step 9: Update the internal map with the newly discovered information.

Step 10: If a previously possible path is found to be blocked, remove it and select another path.

Step 11: Continue this process until the robot reaches G.

Cost Calculation

For a normal cell:

Cost = 1

For a risky cell:

Cost = 1 + 2 = 3

The total path cost is the sum of the costs of all movements.

For example, if a path contains:

- 4 normal movements

- 1 risky movement

Then:

Total cost = (4 × 1) + (1 × 3)

Total cost = 7

The robot should select a path with the minimum total cost while maintaining safe navigation.

c) Application of AI Concepts

1. Intelligent Agent

The rescue robot acts as an intelligent agent.

It receives information from the environment through its sensors and takes actions based on that information.

The basic process is:

Environment → Sensors → Robot/Agent → Decision → Action

For example, if the robot detects an obstacle in front of it, it can decide to turn left or right instead of moving into the blocked cell.

Thus, the robot perceives the environment and performs suitable actions to achieve its goal.

2. Rationality

A rational agent chooses an action that is expected to give the best result according to its available information.

Here, the robot's goal is not simply to reach G. It must reach G while minimizing cost and avoiding unsafe areas.

For example, suppose there are two possible paths:

- Path A: shorter but contains a risky zone
- Path B: slightly longer but completely safe

If Path B has a lower total cost, the rational robot should choose Path B.

Therefore, rationality means choosing the best available action based on the current knowledge.

3. Environment Type

The environment has several important characteristics.

Partiallyobservable:

The robot cannot see the complete building. It can only sense nearby cells.

Sequential:

The current decision affects future decisions.

Dynamic:

The robot updates its knowledge as it moves and discovers new information.

Discrete:

The environment consists of separate grid cells and individual movements.

Single-agent:

The robot is the main decision-making agent in this problem.

Therefore, the environment can be described mainly as a partially observable, sequential, dynamic and discrete environment.

d) Justification of the Chosen Approach

Uniform Cost Search is suitable because the movement costs are different.

A normal cell has a cost of 1, whereas entering a risky zone has a total cost of 3. UCS considers the accumulated path cost and therefore can find the minimum-cost path.

However, the robot cannot know the complete environment beforehand because its sensing capability is limited. Therefore, an Online Search Agent is also needed.

The robot can move step by step, observe new cells, update its knowledge and change its plan whenever new obstacles or risky areas are discovered.

Hence, the combination of Uniform Cost Search and Online Search is appropriate because it:

- Handles different movement costs.
- Avoids blocked cells.
- Considers risky zones.
- Updates knowledge dynamically.
- Works with partial information.
- Finds a minimum-cost feasible path based on the information available.

Conclusion

The rescue robot can successfully solve the problem by acting as a rational intelligent agent. It observes the nearby environment, updates its knowledge and selects actions based on the lowest safe cost. Uniform Cost Search helps in minimizing the total path cost, while online search allows the robot to adapt when it discovers new information.

Therefore, this approach is suitable for a disaster environment where complete information is not available in advance.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided